

THEORY AND METHODS

THE CENTRAL AND NON-CENTRAL MATRIX VARIATE DIRICHLET TYPE III DISTRIBUTION

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Summary: Let $S_i \sim W_p(n_i, \Sigma)$, $i = 1, \dots, r$ and $B \sim W_p(m, \Sigma, \Theta)$. Consider the random matrices $W_i = S^{-\frac{1}{2}} S_i S^{-\frac{1}{2}}$, $i = 1, \dots, r$, where $S = \sum_{i=1}^r S_i + cB$ and c is an additional parameter. The joint distribution of $W_1, \dots, W_r$ is called the central ($\Theta = \mathbf{0}$) and non-central matrix variate Dirichlet type III distribution. These matrix variate Dirichlet type III distributions may serve as an alternative to the matrix variate Dirichlet type I in multivariate analysis. The marginal property of the central matrix variate Dirichlet type III distribution is derived.

1. Introduction

The methods of multivariate analysis depend upon the multivariate normal distribution. If a sample is drawn from a multivariate normal distribution, the sample covariance matrix has a Wishart distribution. Therefore, this distribution plays an important role in inferential procedures. The matrix variate Beta distribution is closely connected to the Wishart distribution.

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Several test statistics in multivariate analysis of variance and covariance are functions of the Beta matrix. An extension of the matrix variate Beta distribution is the matrix variate Dirichlet distribution which is also important in multivariate statistical analysis (Troskie, 1972).

The matrix Dirichlet type I and II distributions are well known in literature (Olkin and Rubin, 1964 and Troskie, 1967). The matrix variate Dirichlet type III distribution is an extension of the matrix variate Beta type III distribution and it is a more general form of the matrix variate Dirichlet type I distribution.

In Section 2 the non-central matrix Beta type III distribution is defined which is a special case of the non-central matrix variate Dirichlet type III distribution. Some known distributions and their relations are also discussed in this section. In Section 3 the central and non-central matrix variate Dirichlet type III distributions are derived from independent Wishart distributed matrix variables. The marginal property of the central matrix variate Dirichlet type III distribution is derived in Section 4 and it is shown that this property differs from the matrix variate Dirichlet type I and type II distributions. In Section 5 the role of the extra parameter c is discussed for the bivariate case. The effect of this parameter on the correlation between the two random variables is shown and its importance in Bayesian analysis is discussed. Finally, in the appendix we give some known results that are used in this article.

2. Some useful distributions

If $\mathbf{x}_1, \dots, \mathbf{x}_m$ are independent $N_p(\boldsymbol{\mu}, \Sigma)$, then matrix variate $X = (\mathbf{x}_1, \dots, \mathbf{x}_m) \sim N_{p,m}(M, \Sigma \otimes I_m)$ and $B = XX'$ is said to have the *non-central Wishart* distribution, denoted by $W_p(m, \Sigma, \Theta)$ with

$M = \mu \mathbf{1}'$ and $\Theta = \Sigma^{-1} M M'$. The density function of B is given by

$$\left\{ 2^{\frac{1}{2}mp} \Gamma_p \left(\frac{m}{2} \right) \det(\Sigma)^{\frac{1}{2}m} \right\}^{-1} \text{etr} \left(-\frac{1}{2} \Theta \right) \text{etr} \left(-\frac{1}{2} \Sigma^{-1} B \right) \\ \cdot \det(B)^{\frac{1}{2}(m-p-1)} {}_0F_1 \left(\frac{m}{2}; \frac{1}{4} \Theta \Sigma^{-1} B \right), \quad B > 0, m \geq p.$$

If $S \sim W_p(n, \Sigma)$, independent of B , then for $c > 0$, $W = (S + cB)^{-\frac{1}{2}} S (S + cB)^{-\frac{1}{2}}$ is said to have a *non-central matrix variate Beta type III* distribution, denoted as $W \sim B_p^{III} \left(\frac{n}{2}, \frac{m}{2}, c, \Theta \right)$. If $\Sigma = I_p$ then the density function of W is defined as

$$\left\{ \beta_p \left(\frac{n}{2}, \frac{m}{2} \right) \right\}^{-1} c^{\frac{1}{2}np} \det(W)^{\frac{1}{2}n - \frac{1}{2}(p+1)} \det(I_p - W)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \\ \cdot \det[I_p + (c-1)W]^{-\frac{1}{2}(m+n)} \text{etr} \left(-\frac{1}{2} \Theta \right) \\ \cdot {}_1F_1 \left(\frac{m+n}{2}; \frac{m}{2}; \frac{1}{2} (I_p - W)^{\frac{1}{2}} [I_p + (c-1)W]^{-1} (I_p - W)^{\frac{1}{2}} \Theta \right), \quad (1)$$

$0 < W < I_p$, where $\beta_p \left(\frac{n}{2}, \frac{m}{2} \right) = \frac{\Gamma_p \left(\frac{m}{2} \right) \Gamma_p \left(\frac{n}{2} \right)}{\Gamma_p \left(\frac{m+n}{2} \right)}$ and ${}_1F_1$ is the confluent

hypergeometric function of matrix argument.

The following are special cases of the above density:

1. If $\Theta = \mathbf{0}$ then W has the *central matrix variate Beta type III* distribution, i.e. $W \sim B_p^{III} \left(\frac{n}{2}, \frac{m}{2}, c \right)$.
2. The case where $\Theta = \mathbf{0}$ and $c = 2$, that is $W \sim B_p^{III} \left(\frac{n}{2}, \frac{m}{2}, 2 \right)$, is discussed by Gupta and Nagar (2000a). The density of W is then given by

$$\left\{ \beta_p \left(\frac{n}{2}, \frac{m}{2} \right) \right\}^{-1} 2^{\frac{1}{2}np} \det(W)^{\frac{1}{2}n - \frac{1}{2}(p+1)} \det(I_p - W)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \\ \cdot \det(I_p + W)^{-\frac{1}{2}(m+n)}.$$

3. When $\Theta = \mathbf{0}$ and $c = 1$, then W has the well-known *matrix variate Beta type I* distribution, i.e. $W \sim B_p^I \left(\frac{n}{2}, \frac{m}{2} \right)$.

The introduction of the additional parameter c , gives a more parameter-rich distribution compared to the matrix variate Beta type I distribution. The need for a parameter-rich family of multivariate distributions has been expressed in literature in order to accommodate prior information of the decision maker (Bekker, 1990; Press, 1982).

3. Central and non-central matrix variate Dirichlet type III distribution

In the following theorem we derived the matrix variate Dirichlet type III distribution.

Theorem 1

Let $S_i \sim W_p(n_i, \Sigma)$, $i = 1, \dots, r$ and $B \sim W_p(m, \Sigma, \Theta)$ be independently distributed. Define

$$W_i = S^{-\frac{1}{2}} S_i S^{-\frac{1}{2}}, \quad i = 1, \dots, r,$$

where $S = \sum_{i=1}^r S_i + cB$, $c > 0$, $S = S^{\frac{1}{2}} S^{\frac{1}{2}}$, $0 < W_i < I_p$ and $0 < \sum_{i=1}^r W_i < I_p$, $i = 1, \dots, r$. Furthermore, $n_i > (p-1)$, $i = 1, \dots, r$ and $m > (p-1)$.

Let $n = \sum_{i=1}^r n_i$.

Consider the following three cases. The first two give the density of the non-central matrix variate Dirichlet III distribution and the third case gives the density of the central matrix variate Dirichlet III distribution.

1. The joint density of $(W_1, \dots, W_r)$ is

$$\begin{aligned} & \left[2^{\frac{1}{2}(m+n)p} \Gamma_p\left(\frac{m}{2}\right) \prod_{i=1}^r \Gamma_p\left(\frac{n_i}{2}\right) \det(\Sigma)^{\frac{1}{2}(m+n)} \right]^{-1} c^{-\frac{1}{2}mp} \text{etr}\left(-\frac{1}{2}\Theta\right) \\ & \cdot \prod_{i=1}^r \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \det\left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \\ & \cdot \int_{S>0} \text{etr}\left[-\frac{1}{2c}\Sigma^{-1}S^{\frac{1}{2}}\left(I_p + (c-1)\sum_{i=1}^r W_i\right)S^{\frac{1}{2}}\right] \det(S)^{\frac{1}{2}(m+n) - \frac{1}{2}(p+1)} \\ & \cdot {}_0F_1\left(\frac{m}{2}; \frac{1}{4c}\Theta\Sigma^{-1}S^{\frac{1}{2}}\left(I_p - \sum_{i=1}^r W_i\right)S^{\frac{1}{2}}\right) dS. \end{aligned} \quad (2)$$

2. If $\Sigma = I_p$ then the joint density of $(W_1, \dots, W_r)$ is

$$\begin{aligned} & \left\{ \beta_p\left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}\right) \right\}^{-1} c^{\frac{1}{2}np} \prod_{i=1}^r \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\ & \cdot \det\left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \det\left(I_p + (c-1)\sum_{i=1}^r W_i\right)^{-\frac{1}{2}(m+n)} \text{etr}\left(-\frac{1}{2}\Theta\right) \\ & \cdot {}_1F_1\left(\frac{m+n}{2}; \frac{m}{2}; \frac{1}{2}\left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}}\left(I_p + (c-1)\sum_{i=1}^r W_i\right)^{-1}\left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}}\Theta\right), \end{aligned} \quad (3)$$

$$\text{where } \beta_p\left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}\right) = \frac{\prod_{i=1}^r \Gamma_p\left(\frac{n_i}{2}\right) \Gamma_p\left(\frac{m}{2}\right)}{\Gamma_p\left(\frac{m+n}{2}\right)}.$$

3. If $\Sigma = I_p$ and $\Theta = \mathbf{0}$ then the density of $(W_1, \dots, W_r)$ is

$$\begin{aligned} & \left\{ \beta_p\left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}\right) \right\}^{-1} c^{\frac{1}{2}np} \prod_{i=1}^r \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\ & \cdot \det\left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \det\left(I_p + (c-1)\sum_{i=1}^r W_i\right)^{-\frac{1}{2}(m+n)}. \end{aligned} \quad (4)$$

This is denoted as $(W_1, \dots, W_r) \sim D_p^{III}\left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}; c\right)$.

Proof:

The joint density of $(S_1, \dots, S_r, B)$ is given by

$$K \prod_{i=1}^r \left[\text{etr} \left(-\frac{1}{2} \Sigma^{-1} S_i \right) \det(S_i)^{\frac{1}{2}(n_i - p - 1)} \right] \\ \cdot \text{etr} \left(-\frac{1}{2} \Theta \right) \text{etr} \left(-\frac{1}{2} \Sigma^{-1} B \right) \det(B)^{\frac{1}{2}(m - p - 1)} {}_0F_1 \left(\frac{m}{2}; \frac{1}{4} \Theta \Sigma^{-1} B \right) \quad (5)$$

where $K^{-1} = 2^{\frac{1}{2}(m+n)p} \Gamma_p \left(\frac{m}{2} \right) \prod_{i=1}^r \Gamma_p \left(\frac{n_i}{2} \right) \det(\Sigma)^{\frac{1}{2}(m+n)}$ and $n = \sum_{i=1}^r n_i$.

Making the transformation $S = \sum_{i=1}^r S_i + cB$, $W_i = S^{-\frac{1}{2}} S_i S^{-\frac{1}{2}}$, $i = 1, \dots, r$ with Jacobian

$J(S_1, \dots, S_r, B \rightarrow W_1, \dots, W_r, S) = \det(S)^{\frac{1}{2}r(p+1)} c^{-\frac{1}{2}p(p+1)}$ in (5) we get the joint density

of $(W_1, \dots, W_r, S)$ as

$$f(W_1, \dots, W_r, S) \\ = K c^{-\frac{1}{2}mp} \text{etr} \left(-\frac{1}{2} \Theta \right) \prod_{i=1}^r \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \det \left(I_p - \sum_{i=1}^r W_i \right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \\ \cdot \text{etr} \left[-\frac{1}{2c} \Sigma^{-1} S^{\frac{1}{2}} \left(I_p + (c-1) \sum_{i=1}^r W_i \right) S^{\frac{1}{2}} \right] \det(S)^{\frac{1}{2}(m+n) - \frac{1}{2}(p+1)} \\ \cdot {}_0F_1 \left(\frac{m}{2}; \frac{1}{4c} \Theta \Sigma^{-1} S^{\frac{1}{2}} \left(I_p - \sum_{i=1}^r W_i \right) S^{\frac{1}{2}} \right). \quad (6)$$

Integrating (6) with respect to S gives (2).

Consider the case where $\Sigma = I_p$. Making the transformation $W_i \rightarrow HW_iH'$, $i = 1, 2, \dots, r$ and $S \rightarrow HSH'$ where H ($p \times p$) is orthogonal (Díaz-García and Gutiérrez-Jáimez, 2006) and substituting in (6) gives

$$\begin{aligned}
& f(HW_1H', \dots, HW_rH', HSH') \\
&= Kc^{-\frac{1}{2}mp} \text{etr}\left(-\frac{1}{2}\Theta\right) \prod_{i=1}^r \det(HW_iH')^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\
&\quad \cdot \det\left(I_p - \sum_{i=1}^r HW_iH'\right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \\
&\quad \cdot \text{etr}\left[-\frac{1}{2c}\left(I_p + (c-1)\sum_{i=1}^r HW_iH'\right)HSH'\right] \det(HSH')^{\frac{1}{2}(m+n) - \frac{1}{2}(p+1)} \\
&\quad \cdot {}_0F_1\left(\frac{m}{2}; \frac{1}{4c}\Theta(HSH')^{\frac{1}{2}}(I_p - \sum_{i=1}^r HW_iH')(HSH')^{\frac{1}{2}}\right) \\
&= Kc^{-\frac{1}{2}mp} \text{etr}\left(-\frac{1}{2}\Theta\right) \prod_{i=1}^r \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \det\left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \\
&\quad \cdot \text{etr}\left[-\frac{1}{2c}\left(I_p + (c-1)\sum_{i=1}^r W_i\right)S\right] \det(S)^{\frac{1}{2}(m+n) - \frac{1}{2}(p+1)} \\
&\quad \cdot {}_0F_1\left(\frac{m}{2}; \frac{1}{4c}\Theta\left(HS^{\frac{1}{2}}H'\right)(I_p - \sum_{i=1}^r HW_iH')\left(HS^{\frac{1}{2}}H'\right)\right).
\end{aligned}$$

We consider the symmetrized density function of $(W_1, \dots, W_r)$ defined by Greenacre (1973), that is

$$f_S(W_1, \dots, W_r) = \int_{S>0} \int_{O(p)} f(HW_1H', \dots, HW_rH', HSH') dH dS$$

where H ($p \times p$) is orthogonal and dH is the normalized invariant measure on $O(p)$. Note that $dS = dHSH'$ (Díaz-García and Gutiérrez-Jáimez, 2006).

Then

$$\begin{aligned}
& f_S(W_1, \dots, W_r) \\
&= Kc^{-\frac{1}{2}mp} \text{etr}\left(-\frac{1}{2}\Theta\right) \prod_{i=1}^r \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\
&\quad \cdot \det\left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \int_{S>0} \text{etr}\left[-\frac{1}{2c}\left(I_p + (c-1)\sum_{i=1}^r W_i\right)S\right]
\end{aligned}$$

$$\cdot \det(S)^{\frac{1}{2}(m+n) - \frac{1}{2}(p+1)} \int_{O(p)} {}_0F_1\left(\frac{m}{2}; \frac{1}{4c} H' \Theta H S^{\frac{1}{2}} (I_p - \sum_{i=1}^r W_i) S^{\frac{1}{2}}\right) dH dS. \quad (7)$$

From (A.5) it follows that

$$\begin{aligned} & \int_{O(p)} {}_0F_1\left(\frac{m}{2}; \frac{1}{4c} H' \Theta H S^{\frac{1}{2}} (I_p - \sum_{i=1}^r W_i) S^{\frac{1}{2}}\right) dH \\ &= \int_{O(p)} {}_0F_1\left(\frac{m}{2}; \frac{1}{4c} (I_p - \sum_{i=1}^r W_i)^{\frac{1}{2}} H' \Theta H (I_p - \sum_{i=1}^r W_i)^{\frac{1}{2}} S\right) dH. \end{aligned}$$

Using this, changing the order of integration in (7) and integrating with respect to S using (A.2), gives the symmetrized density, $f_S(W_1, \dots, W_r)$, as

$$\begin{aligned} & \left\{ \beta_p\left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}\right) \right\}^{-1} c^{\frac{1}{2}np} \prod_{i=1}^r \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\ & \cdot \text{etr}\left(-\frac{1}{2}\Theta\right) \det\left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \det\left(I_p + (c-1) \sum_{i=1}^r W_i\right)^{-\frac{1}{2}(m+n)} \\ & \cdot \int_{O(p)} {}_1F_1\left(\frac{m+n}{2}; \frac{m}{2}; \frac{1}{2} \left(I_p + (c-1) \sum_{i=1}^r W_i\right)^{-1} \left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}}\right. \\ & \quad \left. H' \Theta H \left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}}\right) dH. \quad (8) \end{aligned}$$

Since $f_S(W_1, \dots, W_r) = \int_{O(p)} f(HW_1H', \dots, HW_rH') dH$, it follows from (8) that

$$\begin{aligned} & f(HW_1H', \dots, HW_rH') \\ &= \left\{ \beta_p\left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}\right) \right\}^{-1} c^{\frac{1}{2}np} \prod_{i=1}^r \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\ & \cdot \text{etr}\left(-\frac{1}{2}\Theta\right) \det\left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \det\left(I_p + (c-1) \sum_{i=1}^r W_i\right)^{-\frac{1}{2}(m+n)} \\ & \cdot {}_1F_1\left(\frac{m+n}{2}; \frac{m}{2}; \frac{1}{2} \Theta H \left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}} \left(I_p + (c-1) \sum_{i=1}^r W_i\right)^{-1} \left(I_p - \sum_{i=1}^r W_i\right)^{\frac{1}{2}} H'\right) \end{aligned}$$

Making the transformation $HW_iH' \rightarrow W_i$, $i = 1, \dots, r$, gives (3).

The result in part 3 follows immediately from this. ■

Remarks

1. The density of the non-central matrix variate Beta type III distribution defined in (1) can be derived similarly as in part 2 of Theorem 1, with $r = 1$.
2. For $c = 1$, the density in (4) simplifies to that of the matrix variate Dirichlet type I, denoted as $D_p^I\left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}\right)$, (see Gupta and Nagar, 2000b, for detailed discussion). In the multivariate case, when $p = 1$ and $c = 2$, the density is that of the multivariate Dirichlet type III distribution (Cardeño, Nagar and Sánchez, 2005), which is a new member of the multivariate Liouville family.
3. Let $(W_1, \dots, W_r) \sim D_p^{III}\left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}; c\right)$ and $\Psi_1, \dots, \Psi_r, \Omega$ be symmetric matrices such that $\Omega > 0$ and $\Omega - \sum_{i=1}^r \Psi_i > 0$. If Z_i , $i = 1, \dots, r$ is defined as

$$Z_i = \left(\Omega - \sum_{i=1}^r \Psi_i\right)^{\frac{1}{2}} W_i \left(\Omega - \sum_{i=1}^r \Psi_i\right)^{\frac{1}{2}} + \Psi_i,$$

then $(Z_1, \dots, Z_r)$ have the generalized matrix variate Dirichlet type III distribution with density

$$\begin{aligned} & \left\{ \beta_p\left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}\right) \right\}^{-1} c^{\frac{1}{2}np} \det\left(\Omega - \sum_{i=1}^r \Psi_i\right)^{\frac{1}{2}(p+1)} \\ & \cdot \prod_{i=1}^r \det(Z_i - \Psi_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\ & \cdot \det\left(\Omega - \sum_{i=1}^r Z_i\right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \det\left[\Omega - c \sum_{i=1}^r \Psi_i + (c-1) \sum_{i=1}^r Z_i\right]^{-\frac{1}{2}(m+n)}, \\ & \Psi_i < Z_i < \Omega, \quad i = 1, \dots, r, \quad \sum_{i=1}^r Z_i < \Omega. \end{aligned}$$

Bekker et al. (2005) discussed densities of composite generalized matrix variate Beta type I variables, which is a special case of the above when $c = 1$ and $r = 1$.

4. Let $(W_1, \dots, W_r) \sim D_p^{III} \left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}; c \right)$ and define $Z_i = \frac{1}{2}mW_i$, $i = 1, \dots, r$. Then $(Z_1, \dots, Z_r)$ are *asymptotically* distributed as a *product of independent matrix variate gamma densities*, that is

$$\lim_{m \rightarrow \infty} f(Z_1, \dots, Z_r) = \prod_{i=1}^r \frac{\text{etr}(-cZ_i) \det(Z_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \det(cI_p)^{\frac{1}{2}n_i}}{\Gamma_p\left(\frac{n_i}{2}\right)}.$$

4. The marginal property of the central matrix Dirichlet III

In the case of the matrix variate Dirichlet type I and II distributions it is well known that the marginal density is again of the same type and for one matrix variable the density is that of the Beta type I and Beta type II respectively. In Theorem 2 the marginal density of any subset of matrices having a matrix variate Dirichlet type III distribution is derived and it is shown that the above property is not valid for the matrix variate Dirichlet type III distribution.

Theorem 2

If $(W_1, \dots, W_r) \sim D_p^{III} \left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2}; c \right)$, then the joint density of $(W_1, \dots, W_s)$, $s \leq r$ is given by

$$\begin{aligned} & \left\{ \beta_p \left(\frac{n_1}{2}, \dots, \frac{n_s}{2}; \sum_{i=s+1}^r \frac{n_i}{2} + \frac{m}{2} \right) \right\}^{-1} c^{-\frac{1}{2}mp} \prod_{i=1}^s \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\ & \cdot \det \left(I_p - \sum_{i=1}^s W_i \right)^{\sum_{i=s+1}^r \frac{n_i}{2} + \frac{m}{2} - \frac{1}{2}(p+1)} \\ & \cdot {}_2F_1 \left(\frac{m}{2}, \frac{m+n}{2}; \sum_{i=s+1}^r \frac{n_i}{2} + \frac{m}{2}; \frac{c-1}{c} (I_p - \sum_{i=1}^s W_i) \right), \end{aligned} \quad (9)$$

$$0 < W_i < I_p, \quad 0 < \sum_{i=1}^s W_i < I_p \quad \text{and} \quad n = \sum_{i=1}^r n_i.$$

Proof:

The joint density of $(W_1, \dots, W_r)$, given in (4), can be rewritten as

$$\begin{aligned}
& \left\{ \beta_p \left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2} \right) \right\}^{-1} c^{\frac{1}{2}np} \prod_{i=1}^{r-1} \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \det(W_r)^{\frac{1}{2}n_r - \frac{1}{2}(p+1)} \\
& \cdot \det \left(I_p - \sum_{i=1}^{r-1} W_i \right)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \det \left[I_p - \left(I_p - \sum_{i=1}^{r-1} W_i \right)^{-1} W_r \right]^{\frac{1}{2}m - \frac{1}{2}(p+1)} \\
& \cdot \det \left(I_p + (c-1) \sum_{i=1}^{r-1} W_i \right)^{-\frac{1}{2}(m+n)} \\
& \cdot \det \left[I_p + (c-1) \left(I_p + (c-1) \sum_{i=1}^{r-1} W_i \right)^{-1} W_r \right]^{-\frac{1}{2}(m+n)} \quad (10)
\end{aligned}$$

The transformation $Z_r = \left(I_p - \sum_{i=1}^{r-1} W_i \right)^{-1} W_r$ with Jacobian

$J(W_r \rightarrow Z_r) = \det \left(I_p - \sum_{i=1}^{r-1} W_i \right)^{\frac{1}{2}(p+1)}$ in (10) yields the joint density of $(W_1, \dots, W_{r-1}, Z_r)$ as

$$\begin{aligned}
& \left\{ \beta_p \left(\frac{n_1}{2}, \dots, \frac{n_r}{2}; \frac{m}{2} \right) \right\}^{-1} c^{\frac{1}{2}np} \prod_{i=1}^{r-1} \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\
& \cdot \det \left(I_p - \sum_{i=1}^{r-1} W_i \right)^{\frac{1}{2}(n_r+m) - \frac{1}{2}(p+1)} \det \left(I_p + (c-1) \sum_{i=1}^{r-1} W_i \right)^{-\frac{1}{2}(m+n)} \\
& \cdot \det(Z_r)^{\frac{1}{2}n_r - \frac{1}{2}(p+1)} \det(I_p - Z_r)^{\frac{1}{2}m - \frac{1}{2}(p+1)} \\
& \cdot \det \left[I_p + (c-1) \left(I_p + (c-1) \sum_{i=1}^{r-1} W_i \right)^{-1} \left(I_p - \sum_{i=1}^{r-1} W_i \right) Z_r \right]^{-\frac{1}{2}(m+n)} \quad (11)
\end{aligned}$$

The marginal density of $(W_1, \dots, W_{r-1})$ is obtained by integrating out Z_r from (11), by using (A.1) and (A.3) and gives the joint density of $(W_1, \dots, W_{r-1})$ as

$$\begin{aligned} & \left\{ \beta_p \left(\frac{n_1}{2}, \dots, \frac{n_{r-1}}{2}; \frac{n_r+m}{2} \right) \right\}^{-1} c^{-\frac{1}{2}mp} \prod_{i=1}^{r-1} \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\ & \cdot \det \left(I_p - \sum_{i=1}^{r-1} W_i \right)^{\frac{1}{2}(n_r+m) - \frac{1}{2}(p+1)} \\ & \cdot {}_2F_1 \left(\frac{m}{2}, \frac{m+n}{2}; \frac{n_r+m}{2}; \frac{c-1}{c} \left(I_p - \sum_{i=1}^{r-1} W_i \right) \right). \end{aligned}$$

The joint density of $(W_1, \dots, W_{r-1})$ can be rewritten as

$$\begin{aligned} & \left\{ \beta_p \left(\frac{n_1}{2}, \dots, \frac{n_{r-1}}{2}; \frac{n_r+m}{2} \right) \right\}^{-1} c^{-\frac{1}{2}mp} \prod_{i=1}^{r-2} \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \\ & \cdot \det(W_{r-1})^{\frac{1}{2}n_{r-1} - \frac{1}{2}(p+1)} \det \left(I_p - \sum_{i=1}^{r-2} W_i \right)^{\frac{1}{2}(n_r+m) - \frac{1}{2}(p+1)} \\ & \cdot \det \left[I_p - \left(I_p - \sum_{i=1}^{r-2} W_i \right)^{-1} W_{r-1} \right]^{\frac{1}{2}(n_r+m) - \frac{1}{2}(p+1)} \\ & \cdot {}_2F_1 \left(\frac{m}{2}, \frac{m+n}{2}; \frac{n_r+m}{2}; \frac{c-1}{c} \left(I_p - \sum_{i=1}^{r-2} W_i - W_{r-1} \right) \right). \quad (12) \end{aligned}$$

Consider the transformation $Z_{r-1} = \left(I_p - \sum_{i=1}^{r-2} W_i \right)^{-1} W_{r-1}$ with Jacobian

$$J(W_{r-1} \rightarrow Z_{r-1}) = \det \left(I_p - \sum_{i=1}^{r-2} W_i \right)^{\frac{1}{2}(p+1)}.$$

Substituting into (12) yields the joint density of $(W_1, \dots, W_{r-2}, Z_{r-1})$.

Integrating with respect to Z_{r-1} by using (A.4), gives

$$\begin{aligned}
& f(W_1, \dots, W_{r-2}) \\
&= \left\{ \beta_p \left(\frac{n_1}{2}, \dots, \frac{n_{r-2}}{2}, \frac{n_{r-1}+n_r+m}{2} \right) \right\}^{-1} c^{-\frac{1}{2}mp} \\
& \cdot \prod_{i=1}^{r-2} \det(W_i)^{\frac{1}{2}n_i - \frac{1}{2}(p+1)} \det \left(I_p - \sum_{i=1}^{r-2} W_i \right)^{\frac{1}{2}(n_{r-1}+n_r+m) - \frac{1}{2}(p+1)} \\
& \cdot {}_2F_1 \left(\frac{m}{2}, \frac{m+n}{2}, \frac{n_{r-1}+n_r+m}{2}; \frac{c-1}{c} \left(I_p - \sum_{i=1}^{r-2} W_i \right) \right)
\end{aligned}$$

The result follows from induction. ■

5. Role of the parameter c

In this section the role of the parameter c in the density given by (4) is considered for the *bivariate* case, that is when $r = 2$ and $p = 1$. The density in (4) simplifies to

$$\begin{aligned}
f(w_1, w_2) &= \frac{\Gamma \left(\frac{n_1+n_2+m}{2} \right)}{\Gamma \left(\frac{n_1}{2} \right) \Gamma \left(\frac{n_2}{2} \right) \Gamma \left(\frac{m}{2} \right)} c^{\frac{1}{2}(n_1+n_2)} w_1^{\frac{1}{2}n_1-1} w_2^{\frac{1}{2}n_2-1} \\
& \cdot (1-w_1-w_2)^{\frac{1}{2}m-1} [1+(c-1)w_1+(c-1)w_2]^{-\frac{1}{2}(n_1+n_2+m)}, \quad (13)
\end{aligned}$$

and is denoted as $(W_1, W_2) \sim D^{III} \left(\frac{n_1}{2}, \frac{n_2}{2}; \frac{m}{2}; c \right)$. For $c = 1$ the density in (13) is that of the Dirichlet type I distribution.

The expression for the marginal density of w_1 in (9) simplifies to

$$\begin{aligned}
f(w_1) &= \frac{\Gamma \left(\frac{n_1+n_2+m}{2} \right)}{\Gamma \left(\frac{n_1}{2} \right) \Gamma \left(\frac{n_2+m}{2} \right)} c^{-\frac{1}{2}m} w_1^{\frac{1}{2}n_1-1} (1-w_1)^{\frac{1}{2}(n_2+m)-1} \\
& \cdot {}_2F_1 \left(\frac{m}{2}, \frac{n_1+n_2+m}{2}; \frac{n_2+m}{2}; \frac{c-1}{c} (1-w_1) \right).
\end{aligned}$$

Figure 1 and Figure 2 show graphs of the densities and marginal densities of a $D^{III}(2, 2; 2; c)$ distribution for different values of c . As c increases

the density shifts towards smaller values of the variables. Figure 3 shows the correlation between the two variables W_1 and W_2 for increasing c . The correlation coefficient was calculated for different values of c by using the product moment of W_1 and W_2 associated with (13), that is

$$E \left[W_1^{h_1} W_2^{h_2} \right] = \frac{\Gamma \left(\frac{n_1}{2} + h_1 \right) \Gamma \left(\frac{n_2}{2} + h_2 \right)}{\Gamma \left(\frac{n_1}{2} \right) \Gamma \left(\frac{n_2}{2} \right)} \frac{\Gamma \left(\frac{n_1 + n_2 + m}{2} \right)}{\Gamma \left(\frac{n_1 + n_2 + m}{2} + h_1 + h_2 \right)} c^{\frac{1}{2}(n_1 + n_2)} \cdot {}_2F_1 \left(\frac{n_1 + n_2}{2} + (h_1 + h_2), \frac{n_1 + n_2 + m}{2}; \frac{n_1 + n_2 + m}{2} + (h_1 + h_2); 1 - c \right).$$

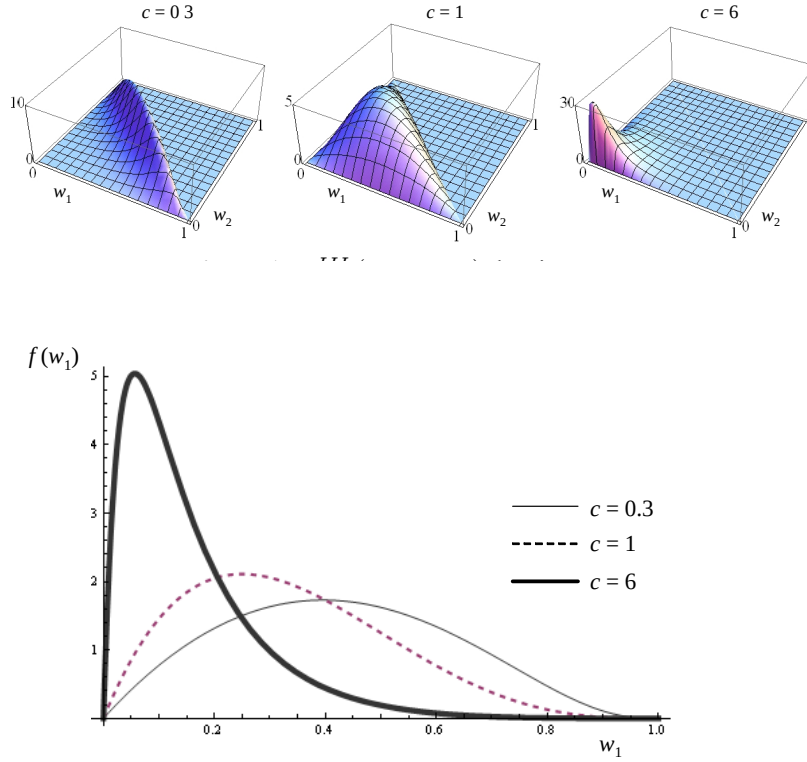


Figure 2 Marginal density, $f(w_1)$, of $D^{III}(2, 2; 2; c)$ density

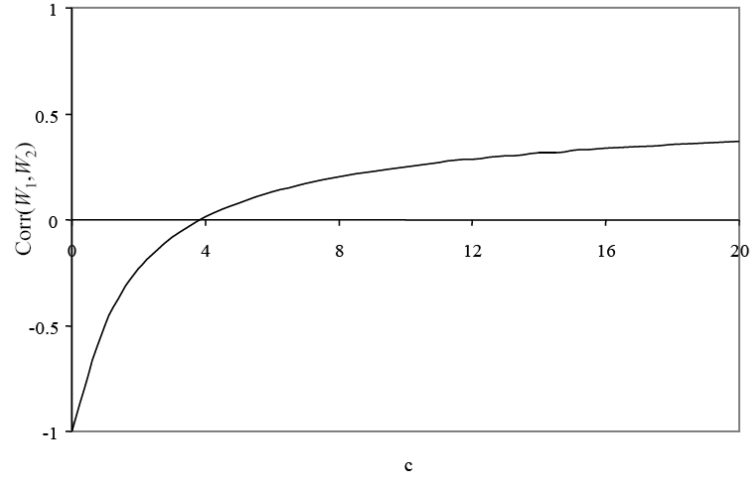


Figure 3 Effect of c on $\text{Corr}(W_1, W_2)$, $(W_1, W_2) \sim D^{III}(2, 2; 2; c)$

The Dirichlet type I distribution is used in Bayesian analysis as the natural conjugate prior for the multinomial distribution where the variables are negatively correlated. In some practical cases random variables may be positively correlated, hence the Dirichlet type I distribution will not be a reasonable choice to be a prior distribution. However, the Dirichlet type III accommodates positive correlation for specific choices of the parameter c , as shown in Figure 3 for the bivariate case.

6. Conclusion

In this paper we give the matrix variate Beta type III distribution and extend it to the matrix variate Dirichlet type III distribution. In the case of the central matrix variate Dirichlet type III distribution we showed that the marginal distribution is not again of the same type. The marginal property of a non-central matrix variate Dirichlet type III will receive attention in future research.

We have also given an alternative to the Dirichlet type I distribution used as a prior for the parameters of the multinomial distribution in Bayesian analysis. The Bayesian estimation of Shannon entropy using this model as a prior will be discussed in a follow-up paper.

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Appendix

The definition of the hypergeometric function of matrix argument and some lemmas which are used in this article follows:

Definition (Greenacre, 1972, Definition 2.3.1, page 11)

The hypergeometric function of matrix argument is defined by

$${}_mF_n(a_1, \dots, a_m; b_1, \dots, b_n; S) = \sum_{k=0}^{\infty} \sum_{\kappa} \frac{(a_1)_{\kappa} \cdots (a_m)_{\kappa}}{(b_1)_{\kappa} \cdots (b_n)_{\kappa}} \frac{C_{\kappa}(S)}{k!},$$

where $a_i, i = 1, \dots, m; b_j, j = 1, \dots, n$ are arbitrary numbers, $S (p \times p)$ is a real symmetric matrix and $\sum_{\kappa}$ denotes summation over all partitions κ .

The generalized hypergeometric coefficient $(a)_{\kappa}$ is given by

$$(a)_{\kappa} = \prod_{i=1}^m \left(a - \frac{1}{2}(i-1)\right)_{k_i},$$

where $(a)_k = a(a-1)\dots(a+k-1)$, $(a)_0 = 1$. Furthermore $C_\kappa(S)$ is the zonal polynomial of S .

Lemma 1 (On the analogy of Gupta and Nagar, 2000b, Equation 1.6.8, page 36)

For $\gamma > \frac{1}{2}(p-1)$, $(\gamma - \alpha) > \frac{1}{2}(p-1)$ and symmetric R ($p \times p$) where $R < I_p$

$$\begin{aligned} & {}_2F_1(\alpha, \beta; \gamma; R) \\ = & \frac{\Gamma_p(\gamma)}{\Gamma_p(\alpha)\Gamma_p(\gamma - \alpha)} \int_{0 < S < I_p} \det(S)^{\alpha - \frac{1}{2}(p+1)} \det(I_p - S)^{\gamma - \alpha - \frac{1}{2}(p+1)} \\ & \cdot \det(I_p - RS)^{-\beta} dS \end{aligned} \quad (\text{A.1})$$

Lemma 2 (Greenacre, 1972, Equation 2.3.5, page 12)

For R ($p \times p$) a positive definite matrix, T ($p \times p$) a symmetric matrix and $t > \frac{1}{2}(p-1)$

$$\begin{aligned} & \int_{S > 0} \text{etr}(-RS) \det(S)^{t - \frac{1}{2}(p+1)} {}_mF_n(a_1, \dots, a_m; b_1, \dots, b_n; ST) dS \\ = & \Gamma_p(t) \det(R)^{-t} {}_{m+1}F_n(a_1, \dots, a_m, t; b_1, \dots, b_n; TR^{-1}). \end{aligned} \quad (\text{A.2})$$

Lemma 3 (Gupta and Nagar, 2000b, Equation 1.6.10, page 37)

$${}_2F_1(\alpha, \beta; \gamma; S) = \det(I_p - S)^{-\beta} {}_2F_1\left(\gamma - \alpha, \beta; \gamma; -S(I_p - S)^{-1}\right). \quad (\text{A.3})$$

Lemma 4 (Greenacre, 1972, Equation 2.3.6, page 13)

Let $R(p \times p)$ be a symmetric matrix, then

$$\begin{aligned} & \int_{0 < S < I_p} \det(S)^{t - \frac{1}{2}(p+1)} \det(I_p - S)^{u - \frac{1}{2}(p+1)} \\ & \cdot {}_m F_n(a_1, \dots, a_m; b_1, \dots, b_n; RS) dS \\ & = \frac{\Gamma_p(t) \Gamma_p(u)}{\Gamma_p(t+u)} {}_{m+1} F_{n+1}(a_1, \dots, a_m, t; b_1, \dots, b_n, t+u; R). \quad (\text{A.4}) \end{aligned}$$

Lemma 5

If X a $p \times p$ symmetric matrix and $A(p \times p) > 0$, $B(p \times p) > 0$, then

$$\int_{O(p)} C_\kappa \left(A^{\frac{1}{2}} B A^{\frac{1}{2}} H X H' \right) dH = \int_{O(p)} C_\kappa \left(B^{\frac{1}{2}} A B^{\frac{1}{2}} H X H' \right) dH. \quad (\text{A.5})$$

Proof:

From Muirhead 1982, equation 36, page 243 and square root factorization it follows that

$$\begin{aligned} \int_{O(p)} C_\kappa \left(A^{\frac{1}{2}} B A^{\frac{1}{2}} H X H' \right) dH &= \frac{C_\kappa \left(A^{\frac{1}{2}} B A^{\frac{1}{2}} \right) C_\kappa(X)}{C_\kappa(I_p)} \\ &= \frac{C_\kappa \left(B^{\frac{1}{2}} A B^{\frac{1}{2}} \right) C_\kappa(X)}{C_\kappa(I_p)} \\ &= \int_{O(p)} C_\kappa \left(B^{\frac{1}{2}} A B^{\frac{1}{2}} H X H' \right) dH. \end{aligned}$$

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